

Real-time detection of fraudulent credit-card transactions using a temporal convolutional autoencoder hybrid model



1 Introduction

Machine learning techniques are used by financial institutions and researchers as a means of detecting credit card fraud. Basic supervised learning algorithms, such as rule based classifiers (i.e. decision trees, random forests) and logistic regression, typically show high levels of accuracy and are highly efficient for this purpose [1]. However, they do have relatively low adaptability compared to more complex algorithms and typically need to be retrained when new iterations of data become available. These limitations, along with well known and common issues with machine learning algorithms such as issues with scalability and high sensitivity to noise within the training data, often make these methods inappropriate for real time credit-card fraud detection [2]. Perhaps a more suitable solution for these issues would be the use of deep learning algorithms. Deep learning algorithms are typically unsupervised learning algorithms and use multi-layered neural networks to analyse data, often by detecting patterns in unstructured data and clustering it accordingly.

1.1 Long-Short Term Memory Networks

A wide variety of deep learning algorithms exist and have already been explored as a part of research into credit-card fraud. Due to the longitudinal nature of financial information, Long Short-Term Memory (LSTM) networks are a popular choice of deep learning algorithm to tackle this problem [1], as they are specifically designed to handle long-term dependencies present in the data. Paying attention to the longitudinal nature of fraudulent financial data has come much more into prominence as research has continued into the characteristics present behind fraudulent transactions, such as

what some researchers call the 'Non-trusted time rule' [3]. This is the idea that fraud is much more likely to occur outside of working hours - this time period is defined in this research as between 8 pm and 6 am. Afriyie et al. [4] found that fraudulent transactions were much more likely to occur between 10pm and 4am than at any other time and posit that this is due to card-holders likely being asleep at this time, as well as banks being closed and unable to monitor transactions. The high accuracy of LSTM networks at the classification of fraudulent transactions in this data provides support for the treatment of this data as a time-series, rather than treating each data point individually with no consideration for the data collected from the same account at an earlier time [5].

Whilst LSTMs are highly accurate at this classification task when conducted appropriately, they also have significant drawbacks. LSTMs are based on recurrent neural networks (RNN), which use internal feedback loops - back-propagation - as a memory mechanism, processing data in sequence and updating its hidden state over time. This process can make their training highly memory intensive, and due to the sequential nature of the processing this cannot be parallelised. Additionally, LSTMs, while designed to avoid the RNN's problem of the vanishing gradient, remain highly prone to exploding gradients. The exploding gradient problem occurs when a gradient is repeatedly multiplied by a large weight matrix during back-propagation, and this usually results in the gradient becoming abnormally large, or NaN.

1.2 Temporal Convolutional Networks

These problems can be avoided by fitting a temporal convolutional network (TCN). TCNs typically perform to similar, or better, rates as LSTMs on these tasks, and are thought to be less computationally expensive [6]. When using a TCN for a time-series prediction, the model uses layers of 1-d convolutions to produce a prediction based on the available data. Two key features of TCNs are typically used in a TCN that is utilised in this manner. Firstly, this model would use causal convolutions. This would result in the output value of the model at any particular time only resulting from data present in the time-series before that time point [7]. This prevents future leakage, which is another issue commonly seen in recurrent neural networks [6]. Secondly, TCNs modelling time-series data typically use dilated convolutions. Dilated convolutions are a technology that increases the receptive field of each layer of the network, without increasing the number of calculations required [7]. Dilated convolutions are particularly necessary in time-series modelling due to the length of data needing to be captured, and stored, at one time by the model. A dilated causal convolution is expressed as follows:

$$y[n] = (x *_{q} h)[n] = \sum_{i=0}^{k-1} h[i]^T x[n - qi]$$

2 Research Questions

Credit-card fraud remains a significant and growing threat to consumers and financial institutions worldwide, resulting in approximately £9.4 billion lost each year [8].

Research regarding detection of fraudulent transactions typically considers machine learning approaches, which are well established and often suitably accurate, but are similarly plagued with methodological challenges. Perhaps the most significant of these challenges is severe class imbalance present in the available data, a well known credit-card fraud dataset where fraudulent transactions make up just 0.172% of data points. These conventional classification based approaches therefore must mitigate class imbalance using synthetic over- or under-sampling methods, such as SMOTE and GAN, which will be covered in more detail in the proposed methodology section. These methods, however, come with their own disadvantages, such as noise generation and over-fitting[9].

This research proposes an alternative framing - to treat credit-card fraud detection as an anomaly detection problem within time-series data rather than as a binary classification task, which will circumvent the need for synthetic minority oversampling. Specifically, it investigates how a temporal convolutional network and auto-encoder hybrid model can be applied to detect fraudulent transactions using reconstruction error from encoding and decoding. The original architecture[10] was developed for use in medical settings, and its use for fraud detection is therefore unexplored. Questions remain surrounding how this model will need to be adapted for this use, including for optimal calibration, error threshold setting, and regarding how the model will perform when compared to conventional classification based methods.

Considering this information, related literature and the proposed methodology set out below, the research questions for this thesis are as follows:

1. To what extent can a hybrid temporal convolutional network - auto-encoder model, trained for anomaly detection, accurately distinguish fraudulent from non-fraudulent credit-card transactions in a severely class-imbalanced dataset?
2. How do architectural choices in the model— such as encoder and decoder depth, kernel size, and the number of convolutional filters — influence the model’s reconstruction accuracy and overall fraud detection performance?
3. How does the use of PCA-transformed features, in place of the original transactional variables, affect the model’s ability to learn the underlying distribution of non-fraudulent behaviour and identify deviations indicative of fraud?
4. How does the anomaly detection approach compare in detection performance — as measured by precision, recall, and F1-score — to conventional classification-based methods that employ synthetic oversampling techniques such as SMOTE?

3 Proposed Methodology

The aim of this thesis is to produce a deep learning model that can be used to detect the likelihood of a credit-card transaction being fraudulent. In order to complete this task, I will be fitting a hybrid Auto-encoder (AE) and TCN model on the well-known Credit Card Fraud detection dataset[11]. This dataset is very widely used in credit-card fraud detection research, and contains just over 284,000 transactions which were collected over two days in September 2013. Of these transactions, 492 (0.172%) are classified as fraudulent, resulting in severe class imbalance - which is an issue often needing to be mitigated in research using this data. The data contains 31 variables, of

which 28 are numeric-only variables previously transformed using principal component analysis (PCA), as well as time, amount spent per transaction, and the fraud target label. It is key here that time is classified as the time elapsed since the first transaction in the dataset, rather than the actual time of each transaction. The original features collected before PCA transformation are not available for research use due to privacy concerns surrounding the data. Several synthetic dataset generators are also available, such as PaySim[12], and can be used to test if there is a significant difference in the efficacy of the model when trained or tested on data with labelled features vs PCA-transformed features, such as those found in the current data.

3.1 Severe Class Imbalance

Severe class imbalance in this dataset is typically tackled using synthetic minority over-sampling techniques, such as Synthetic Minority Over-sampling Technique (SMOTE) or Generative Adversarial Networks (GAN)[9]. Synthetic oversampling techniques such as SMOTE are preferred over techniques like random oversampling because they produce synthetic data points interpolated from the original data, rather than duplicating points in the minority class. Duplicating points in the minority class increases the risk of over-fitting in the model, as it does not consider the underlying distribution of the minority class and can introduce large amounts of noise into the dataset[9, 13]. SMOTE attempts to mitigate this issue by interpolating synthetic data points that fall between the points in the original data. It does, however, remain highly susceptible to noise generation and has issues surrounding data diversity[9]. GAN can be used as a synthetic minority oversampling technique by generating data points that match the underlying distribution of the minority class - rather than just producing points in between data points in the same way as SMOTE[13, 14]. In this case, the severe class imbalance present in the data will be addressed by considering this research as a problem of anomaly detection, rather than as a classification task. When considering an anomaly detection problem using deep learning, class imbalance mitigation methods are not used as the distribution of the minority class is not learned by the model.

3.2 Temporal Convolutional Auto-encoder

This research will use a hybrid temporal convolutional auto-encoder, using adapted architecture from research conducted by Thill et al.[10], for anomaly detection. This model includes an encoder and a decoder layer developed from a temporal convolutional network, which compresses each data point and reconstructs it in output. During the training process, the model will generally attempt to minimise the level of reconstruction error present in output from the majority class (non-fraudulent transactions) as much as possible. The capability for the model to accurately reconstruct the majority class will increase as the underlying distribution of the data is learned. Data not found in this majority class (i.e. fraudulent transactions) will have significantly higher reconstruction error than the rest of the data. To identify data points with high reconstruction error, a distribution of this error will be estimated as the model is trained. Points with a high reconstruction error will be the points with the largest distance from the centre of the distribution - these points would therefore be

classified as an anomaly. As deep learning methods are largely unsupervised learning methods, a small amount of labelled data would then be used to calibrate this model and produce a cut-off value for the reconstruction error. The original research in which this architecture was developed used approximately 10% of their available data for this task[10]. During testing and deployment, any data points with reconstruction error higher than this cut-off value would therefore be classified as fraudulent.

4 Expected Contributions

This research makes several potential contributions to financial fraud detection and the application of deep learning. Firstly, this is a novel application of the hybrid temporal convolutional network auto-encoder hybrid architecture to credit-card fraud detection. Whilst this hybrid model has been previously explored in time-series anomaly detection, particularly in regards to anomaly detection in cardiac monitoring, its application to financial transaction data is currently limited. This thesis extends the architecture proposed by Thill et al. (2021)[10] to this domain, and will provide a concrete evaluation of its appropriateness for detecting fraudulent transactions. Secondly, this research provides an empirical investigation into anomaly detection modelling as an alternative to severe class imbalance mitigation. By framing fraud detection as an anomaly detection problem rather than a classification task, this research offers an evidence-based comparison of the two methods. The findings will contribute to ongoing methodological debate surrounding appropriate approaches to severe class imbalance in fraud detection research.

Thirdly, the structure of the data will hopefully give insights into the effect of PCA-transformed features on unsupervised anomaly detection. Since the dataset’s original features are unavailable due to privacy constraints, using synthetic data as comparison this research will provide analysis of how the opacity and structure of PCA components affect the TCN-AE’s ability to learn non-fraudulent behaviour and flag deviations. This will have practical relevance for deployments in which data is similarly restricted. Finally, this research will provide further guidance into the calibration methods of the current architecture. The TCN-AE model requires small amounts of labelled data to calibrate the reconstruction error distribution and determine a classification threshold. Investigating the amount of data required for this task will provide some practical insight into the minimum labelling requirements for effective deployment in real-world situations. Reducing the amount of labelled fraudulent data necessary to calibrate a model in a deployment scenario significantly decreases the cost of the model, improving its real-world value.

5 Timeline

This research began on 14/04 with consolidation of the existing literature surrounding credit-card fraud detection, appropriate detection methods, and potential methodology. During this time period I shifted to focus on deep learning anomaly detection methods, with a particular interest in TCN-AE architecture. Following on from the submission of this proposal and the assessment period, I will turn to exploring the credit card fraud dataset. This will occur over approximately one week and include

conducting exploratory data analysis, assessing the structure and distribution of the PCA-transformed feature, and any preprocessing required for model input. Over the course of June and early July I will begin developing the model as well as training and evaluation. The TCN-AE model will be adapted from the architecture proposed by Thill et al., and this phase will include adaptations to the encoder and decoder configurations as necessary and to the TCN hyper-parameters, in order to produce a stable model. Training of the model will then occur as set out in the proposed methodology section, in model performance will be evaluated according to the research questions.

Following development and training of the model, the TCN-AE will be compared against classification-based models that typically employ SMOTE or GAN for class imbalance mitigation. Results from these models will be compared against appropriate accuracy measures to assess the appropriateness of each method. Writing of the thesis will occur during all methodological testing, to ensure transparency and completeness, and will conclude in late July, with revisions and edits occurring in August before submission.

References

- [1] Hafez, I.Y., Hafez, A.Y., Saleh, A., Abd El-Mageed, A.A., Abohany, A.A.: A systematic review of ai-enhanced techniques in credit card fraud detection. *Journal of Big Data* **12**(1) (2025) <https://doi.org/10.1186/s40537-024-01048-8>
- [2] Varmedja, D., Karanovic, M., Sladojevic, S., Arsenovic, M., A., A.: Credit Card Fraud Detection - Machine Learning methods. Paper presented at the 18th International Symposium INFOTEH-JAHORINA (2019)
- [3] Xie, Y., Liu, G., Cao, R., Li, Z., Yan, C., Jiang, C.: A Feature Extraction Method for Credit Card Fraud. Paper presented at the 2nd International Conference on Intelligent Autonomous Systems (ICoIAS) (2019)
- [4] Afriyie, J.K., Tawiah, K., Pels, W.A., Addai-Henne, S., Dwamena, H.A., Owiredun, E.O., Ayeh, S.A., Eshun, J.: A supervised machine learning algorithm for detecting and predicting fraud in credit card transactions. *Decision Analytics Journal* (2023)
- [5] Wiese, B., Omlin, C.: Credit card transactions, fraud detection, and machine learning: Modelling time with lstm recurrent neural networks. *Studies in Computational Intelligence*, 231–268 (2009)
- [6] Gopali, S., Abri, F., Siarni-Namini, S., Namin, A.S.: A Comparison of TCN and LSTM Models in Detecting Anomalies in Time Series Data. Paper presented at the 2021 IEEE International Conference on Big Data (2021)
- [7] Wang, C., Liu, Q., Qu, L., Chang, W., Wang, T., Duan, P., Cao, Y.: Self-attention-enhanced Temporal Convolutional Networks for Anomaly Detection in Time Series. Paper presented at the 4th International Conference on Cryptography,

- [8] Team, C.P.: £9.4 billion stolen from UK customers in a year (2025). https://www.cifas.org.uk/newsroom/9.4billion_stolenfromconsumers
- [9] Yang, Y., Khorshidi, H.A., Aickelin, U.: A review on over-sampling techniques in classification of multi-class imbalanced datasets: insights for medical problems. *Frontiers in Digital Health* **6** (2024)
- [10] Thill, M., Konen, W., Wang, H., Back, T.: Temporal convolutional autoencoder for unsupervised anomaly detection in time series. *Applied Soft Computing* (2021)
- [11] ULB, Group, M.L.: Credit Card Fraud Detection. Kaggle (2019)
- [12] Lopez-Rojas, E.A., Elmir, A., Axelsson, S.: PaySim: A financial mobile money simulator for fraud detection. Paper presented at the 28th European Modeling and Simulation Symposium-EMSS (2016)
- [13] Gurcan, F., Soyulu, A.: Synthetic boosted resampling using deep generative adversarial networks: A novel approach to improve cancer prediction from imbalanced datasets. *Cancers* **26**(23), 40–46 (2024) <https://doi.org/10.3390/cancers16234046>
- [14] Joshi, C., Kaloskamps, I., Nolan, L.: Generative adversarial networks (GANs) for synthetic dataset generation with binary classes (2019). <https://datasciencecampus.ons.gov.uk/projects/generative-adversarial-networks-gans-for-synthetic-dataset-generation-with-binary-classes/>